

ICT Lab 6: MS Access – Advanced Student Management System

Objective

Students will design a more advanced database system with multiple conditions, calculated queries, and logical filtering.

Lab Tasks

Task 1 – Create Database (5 minutes)

Create a database named: AdvancedStudentDB.accdb

Task 2 – Create Table (10 minutes)

Create table Students:

StudentID – AutoNumber (Primary Key)

Name – Short Text

Age – Number

Department – Short Text

Semester – Number

GPA – Number

Status – Short Text (Active/Inactive)

Task 3 – Insert Records (10 minutes)

Insert at least 8 records with different departments and GPA values.

Name	Age	Department	Semester	GPA	Status
Ali Khan	19	CS	1	3.2	Active
Sara Ahmed	20	BBA	2	3.5	Active
Ahmed Raza	18	CS	1	2.3	Inactive
Fatima Noor	21	IT	3	3.8	Active
Hassan Ali	19	CS	2	2.4	Active
Ayesha Malik	20	BBA	1	3.0	Inactive
Bilal Hussain	22	CS	4	2.1	Active
Zainab Sheikh	21	IT	2	3.6	Active

Task 4 – Complex Queries (20 minutes)

Create the following queries:

1. Students with GPA > 3.0 AND Department = "CS"
2. Students with GPA < 2.5 OR Semester = 1
3. Students NOT in CS department
4. Create a calculated field:
GradeStatus: "Good" if GPA >= 3, otherwise "Average"

Task 5 – Sorting & Filtering (5 minutes)

Sort students by GPA in descending order.

Filter only Active students.

Task 6 – Form Creation (5 minutes)

Create a form named StudentForm.

Ensure all fields are visible.

Bonus Challenge (Optional – 5 Marks)

Create a query to show:

Students from CS department with GPA between 2.5 and 3.5

Expected Output

- Database with advanced table
- 8+ records
- Complex queries
- Calculated field
- Sorted results
- Functional form